ScopeX (Bumblebee) — Challenges Log & Recommendations

Experiments 1–4 · Compiled from hands-on troubleshooting

1. Purpose

This document logs the practical issues encountered while running Experiments 1–4 (Wheatstone bridge & filter study, digital counter with the 74LS163, and diode I-V / rectifier characterisation) on the ScopeX Advanced Oscilloscope, and separates them into (a) software/hardware issues in ScopeX itself and (b) mismatches between the assignment sheets (originally written for the ADALM1000 + Analog Discovery workflow) and what ScopeX actually offers. Recommendations for both the ScopeX team and future assignment authors are included at the end.

2. Summary Table of Challenges

Experiment	Challenge	Root Cause
Setup / All	SmartScreen blocked app launch	Unsigned executable; not documented in Setup Manual
Week-2 (Bridge/Filter)	No offset/amplitude control on Sine	Sine type only exposes Frequency
Week-2	Loaded CSV gave garbled freq/amplitude	Undocumented CSV format (header, time units)
Week-2	Frequency field had no effect on Arb playback	Arb mode likely uses fixed hardware clock, undocumented
Week-2	CH2 looked flat despite real signal	CH1/CH2 default to different, unmatched V-scales
Week-2	Garbled Freq/Amp readings at slow sample rate	Aliasing; no in-app warning
Week-2	Bode sweep capped near 5 kHz	Manual advertises up to 10 kHz; ceiling undocumented
Digital (74LS163)	Clock line silently inactive	Confused static D1 output with Digital Frequency Generator pin
Digital (74LS163)	No physical pinout reference	Manual lacks a board pinout diagram
Digital (74LS163)	SIG appeared to run but read 0V	Generator chip needs board VCC; undocumented dependency
Experiment 4 (Diode)	No DC source option in Signal Gen	DC listed in manual spec table but absent from UI dropdown
All	Assignments reference ADALM1000, not ScopeX	Sheets not adapted to ScopeX's actual feature set

3. Detailed Discussion

3.1 Installation & Onboarding

- Windows SmartScreen blocked the unsigned Scope-X.exe on first run ('Windows protected your PC'), requiring the 'More info → Run anyway' workaround. This is expected for an unsigned binary but is not mentioned anywhere in the Setup Manual, so a first-time user has no way to know it's safe to override.
- The Windows driver folder (atmel_winusb.inf + supporting DLLs) has no bundled installer executable — the correct install method (right-click → Install on the .inf file, or manual driver update via Device Manager) had to be inferred rather than documented step-by-step in the Setup Manual excerpt available.
- The GitHub repository auto-generates 'Source code (zip/tar.gz)' links on every release, which look identical to the real, named packages (QT-Windows.zip, etc.) but are not usable builds. This is called out in the README but is an easy trap for a first-time downloader.

3.2 Signal Generator — Sine / DC / Offset

- The Sine waveform option only exposes a Frequency field — there is no Amplitude or Offset control, unlike Square (which has a duty-cycle slider). This makes it impossible to directly generate an excitation like $E = 2.5 + 1 \cdot \sin(\omega t)$ using the built-in Sine type.
- The Application Manual's waveform-type list (Sine, Square, Triangle, Sawtooth, DC, Ramp-Up, Ramp-Down) does not match the actual on-screen dropdown (Sin, Square, Tri, RampUp, RampDn, Arb) — notably, DC is documented but not present as a selectable option in the software we used.
- Because there is no DC option, a constant excitation voltage (needed for the diode I-V sweep in Experiment 4) had to be faked by loading a flat-line CSV through Arbitrary Waveform — a workable but non-obvious substitute for a basic, commonly needed function.

3.3 Arbitrary Waveform — Undocumented File Format

- The manual states the preview plot uses a 0–5V voltage axis and a 'normalised' time axis, but does not specify the actual CSV format ScopeX expects (column order, header row present/absent, delimiter, point count limits).
- A first attempt using a CSV with a header row ('Time,Voltage') and real-world time values (seconds) produced a completely garbled result on the scope — wrong frequency, wrong amplitude, and a fixed-notch glitch at the waveform's minimum, with no error message to explain why.
- Removing the header row and switching to normalised time (0–1) fixed the preview plot, but the frequency actually output by the hardware still did not match the value typed into the Frequency field alongside it — suggesting the Frequency control is ignored for Arbitrary-type waveforms and playback speed is instead fixed by the hardware's internal clock and the number of points in the file. This is not documented anywhere.
- A small but consistent 'notch' distortion appeared once per cycle near the waveform's minimum even with a mathematically continuous, periodic CSV — most likely a device-side quantisation/sample-timing artifact rather than a data problem, but there is no way to confirm this without internal documentation.

3.4 Scope Display & Measurement Reliability

- Channel 1 and Channel 2 do not share a default vertical scale. When CH1 was manually zoomed in (e.g., ±2.5V) and CH2 was left at its default ±20V, a perfectly real CH2 signal appeared as a flat line purely due to the scale mismatch — easy to misread as 'no connection' or 'no signal'.
- The Sample Rate must be manually matched to the signal's real frequency. Leaving it at a default that is far too slow for the generated frequency produces a classic aliasing pattern (a slow 'beat' envelope over a fast ripple) rather than a clear error or warning, and the Freq/Amp measurement readouts return nonsensical values in this state (e.g., a signal set to 1 Hz reading as 184 kHz) instead of flagging that the reading is unreliable.
- The Bode Plot sweep could not be extended above roughly 5 kHz in practice, despite the Specifications table advertising a default sweep range of 100 Hz–10 kHz and describing Start/End Frequency as freely user-defined. This ceiling is not documented and was only discovered by trial and error while trying to verify a calculated 16.1 kHz corner frequency.

3.5 Digital I/O vs. Digital Frequency Generator

- The Digital tab has two separate, similarly-located features — the static Digital I/O output toggles (D1–D4, manually set H/L) and the independent Digital Frequency Generator (Start Freq Gen) — and it is not obvious from the UI which physical pin each one drives. In our case, a clock line was mistakenly wired to a static D1 output (which was just sitting at a fixed HIGH) instead of the actual frequency generator pin, and the only symptom was 'the counter isn't counting', with no indication of which of the two features was actually in use.
- There is no pinout diagram for the physical ScopeX board in the manual excerpts available to us, so mapping logical signals (SIG, CH1, CH2, GND, digital generator output, VCC/GND rails) to physical header pins required guesswork and photos rather than a reference diagram.
- It was not documented that the analog SIG generator's own chip may require the board's own VCC to be connected before it will output a real signal — without this, SIG appeared to run correctly in the software

(waveform shown, no errors) while actually outputting ~0V, which is a very hard failure mode to diagnose from the software alone.

3.6 Assignment ↔ Tooling Mismatch

- All four assignment sheets are written against the Analog Discovery / ADALM1000 workflow (e.g., 'Analog Board (ADALM 1000)', '2.5V DC offset built into the board') rather than ScopeX, so several steps (like the Part 2 rectifier's Vdc source, or the DC excitation source in Experiment 4) needed re-interpretation for ScopeX's actual capabilities rather than a direct one-to-one translation.
- The Week-2 excitation formula ($E = 2.5 + 1 \cdot \sin(\omega t)$, later $3.3 + 1 \cdot \sin(\omega t)$) assumes a signal generator with native amplitude+offset controls, which ScopeX's Sine type does not currently provide — this single mismatch cascaded into most of the Arbitrary Waveform and CSV-format issues logged in 3.3.

4. Recommendations for Bumblebee-ScopeX

- Add native Amplitude and Offset fields to the Sine waveform type (matching what Square already has for duty cycle), so common excitations like DC + AC sine do not require the Arbitrary Waveform workaround.
- Add back a true DC output type (as already listed in the Application Manual's own capability table) with a single adjustable voltage field.
- Publish the exact Arbitrary Waveform CSV specification (column order, header yes/no, time units — normalised vs. absolute — point-count limits) directly in the Application Manual, ideally with a sample file.
- Clarify, in the manual and/or in-app, whether/how the Frequency field affects Arbitrary-type playback, and if it currently does nothing, either make it functional or grey it out with a tooltip explaining that playback rate is fixed for this mode.
- Default Channel 1 and Channel 2 to the same vertical scale when both are enabled (or clearly flag scale mismatches), so a real signal on an unscaled channel doesn't visually read as 'no connection'.
- Add an on-screen aliasing warning (or auto-adjust Sample Rate) when the measured/expected signal frequency is a large fraction of the current sample rate, since the current failure mode (garbled frequency/amplitude numbers with no flag) is very hard for a student to self-diagnose.
- Correct the Specifications Summary table's Bode sweep range (or fix the underlying ceiling) so the documented 100 Hz–10 kHz default range matches what the hardware can actually sweep.
- Add a labeled pinout diagram of the physical ScopeX board to the manual, showing SIG, CH1, CH2, GND, VCC/+5V, the Digital Frequency Generator output, and D1–D4, so students don't have to reverse-engineer pin mapping from photos.
- Visually distinguish the Digital I/O static toggles from the Digital Frequency Generator in the UI (e.g., separate panels or a clear 'source' label on each physical pin) to prevent the D1-vs-clock-pin confusion described in 3.5.
- Document, in the Getting Started or Troubleshooting section, that the analog signal generator chip requires the board's own VCC before SIG will output a real signal — this is currently a silent failure mode.
- Update the Application Manual's screenshots and waveform-type list to match the current software build (the manual's Generate Signal + Start Signal Generator two-button flow and Sine/Square/Triangle/Sawtooth/DC list do not match the installed app's single Run Signal Generator button and Sin/Square/Tri/RampUp/RampDn/Arb list).

5. Recommendations for Assignment Redesign

- Rewrite ADALM1000-specific language (e.g., 'Analog Board (ADALM 1000)', assumed 2.5V DC offset) to be tool-agnostic, or provide a short ScopeX-specific appendix per experiment describing the equivalent step in ScopeX's actual UI.
- For excitation signals that combine a DC offset with an AC component (Week-2's $E = 2.5/3.3 + 1 \cdot \sin(\omega t)$), either specify the Arbitrary Waveform CSV approach explicitly as the intended method on ScopeX, or hold off on requiring offset-sine excitations until ScopeX's Sine type supports Amplitude/Offset natively.

- For the Bode/frequency-sweep based experiments (Week-2 Part 3), cap the required sweep range at a value confirmed to work on ScopeX hardware (e.g., ≤5 kHz) rather than assuming the manual's advertised 10 kHz ceiling, or explicitly instruct students to report 'no attenuation observed within instrument range' as a valid, expected outcome when the calculated corner frequency exceeds it.
- For the digital experiment (74LS163), provide an explicit ScopeX pin-mapping table (CLK → Digital Frequency Generator pin, VCC/GND → fixed rails, QA–QD → Digital Input pins or CH1/CH2) instead of leaving pin translation to the student, given the ADALM-to-ScopeX pin names don't correspond directly.
- For Experiment 4's diode I-V sweep, replace 'vary Vs1 using the Analog Board' with an explicit instruction to use ScopeX's Arbitrary-Waveform-as-DC-source workaround (or wait for a native DC output), since a direct multimeter-adjustable DC supply is assumed by the original wording but isn't how ScopeX currently works.
- Consider adding a brief 'known ScopeX limitations' box to each assignment sheet (measurement aliasing at mismatched sample rates, Bode sweep ceiling, no native offset-sine) so students recognize these as expected tool behavior rather than assuming their own wiring or understanding is at fault.

6. Overall Takeaway

Across all four experiments, the majority of time lost was not due to conceptual misunderstanding of the circuits themselves, but to translating assignment language written for a different instrument (ADALM1000) into ScopeX's current feature set, and to a handful of undocumented ScopeX behaviors (Arbitrary Waveform's real CSV format and frequency handling, the Bode sweep ceiling, and the SIG-needs-VCC dependency) that had to be discovered through trial and error rather than reference material. Closing the documentation gaps in Section 4 and aligning the assignment wording per Section 5 would likely eliminate most of the friction logged here for future students.